DUAL LDR LIGHT SENSING SYSTEM

with Relay, Buzzer and RGB LED Control

Component Reference & Technical Guide | Project Level 2.6

> ⚠️ **Real Build Note**
> This project was also built physically using the HW-482 relay module. The LU-5-R relay described throughout this document is the relay component found inside the HW-482 module. The HW-482 module provides the relay on a PCB with screw terminals, status LED, and flyback diode already integrated. All electrical specifications remain identical.

1. Component Overview

Component	Value / Model	Qty	Function
LDR	GL5528 or similar	2	Senses ambient light level; resistance changes with illumination
Resistor — voltage divider	10 kΩ	2	Forms voltage divider with each LDR; sets the midpoint voltage
NPN Transistor	2N2222	1	Amplifies LDR1 signal to drive relay coil (LDR1 channel only)
Resistor — base	1 kΩ	1	Limits base current into transistor; prevents damage, ensures saturation
Relay module	HW-482 (LU-5-R coil, 9V)	1	Electromechanical switch; closes NO contact when coil is energised
Flyback diode	1N4148	1	Suppresses back-EMF voltage spike when relay coil de-energises
RGB LED module	Common cathode, internal R	1	RED pin: analog brightness via LDR1. GREEN pin: analog via LDR2
Active buzzer	5–12V DC	1	Sounds when relay NO contact closes (LDR1 channel)
9V battery	PP3 / 6LR61	1	Powers the entire circuit

2. Component Technical Details

LDR — Light Dependent Resistor
Passive photoresistor | Qty: 2 | Real measured values used in this project

What it is: A resistor whose resistance decreases as light intensity increases. Made from cadmium sulphide (CdS), which releases electrons when photons strike its surface — reducing resistance.

How it works here: Each LDR is the lower arm of a voltage divider (9V → 10kΩ → LDR → GND). In darkness the resistance is high, pulling the midpoint voltage up. In bright light the resistance drops, pulling midpoint voltage down.

Real measured values: Dark = 108,000 Ω (108 kΩ) | Bright light = 506 Ω

Resulting midpoint voltages: Dark ≈ 8.24 V (transistor turns ON) | Bright light ≈ 0.43 V (transistor stays OFF)

Key property: No polarity — can be connected either way. Response is non-linear (logarithmic with light intensity).

Why two LDRs: LDR1 controls the relay/buzzer/red LED channel. LDR2 controls the RGB green channel independently.

Voltage Divider — 10 kΩ Resistor + LDR
Core sensing circuit | Resistor on top, LDR on bottom (v2 configuration)

What it is: Two resistors in series between supply and ground. The junction (midpoint) produces a voltage proportional to the ratio of the two resistances.

Configuration in this project: 9V → 10kΩ (fixed) → Node → LDR (variable) → GND

Behaviour: When LDR resistance is HIGH (dark) → midpoint voltage is HIGH (~8.24V). When LDR resistance is LOW (bright light) → midpoint voltage is LOW (~0.43V). This is the OPPOSITE of a top-LDR configuration.

Why 10 kΩ: Chosen to be comparable to the LDR's mid-range resistance (~10 kΩ in indoor light), giving the widest usable voltage swing across typical lighting conditions.

Analog output: The midpoint voltage changes continuously and smoothly — not just ON/OFF. This is used directly to drive the RGB LED pins for smooth brightness control.

LDR2 channel: Uses the same divider but the midpoint feeds only the RGB GREEN pin — no transistor or relay involved.

NPN Transistor — 2N2222
Bipolar Junction Transistor | Used as a switch, not an amplifier | LDR1 channel only

What it is: A three-terminal semiconductor device. Terminals: Base (B), Collector (C), Emitter (E). A small current at the Base controls a much larger current between Collector and Emitter.

Why it is needed: The LDR1 midpoint can only supply a few milliamps. The relay coil needs 72 mA to energise. The transistor bridges this gap — a small base current (7.54 mA) switches a large collector current (70.4 mA).

Operating mode — saturation: When base current is sufficient, the transistor is fully ON (saturated). V_{CE} drops to ~0.2V. Collector and emitter act like a closed switch. The relay coil is pulled to ground through this switch.

Operating mode — cut-off: When $V_{mid} < 0.7V$ (bright light, $V_{mid} \approx 0.43V$), no base current flows. Transistor is fully OFF. Relay coil remains de-energised.

Pin orientation (2N2222, flat face toward you): Left = Emitter | Middle = Base | Right = Collector. Getting this wrong is the most common wiring error.

Key specs: Max I_C = 600 mA | Max V_{CEO} = 40V | h_{FE} (min) = 100 | V_{BE} = 0.7V

Relay — LU-5-R (inside HW-482 module)

Electromechanical switch | 9V coil | Real build uses HW-482 PCB module

What it is: An electromagnetic switch. When current flows through the coil, it creates a magnetic field that physically moves a metal arm, closing (or opening) a separate set of contacts. This allows a low-current control circuit to switch a higher-current load circuit.

HW-482 module note: In the real physical build, the HW-482 relay module was used. This module contains the LU-5-R relay on a PCB with screw terminals for easy wiring, a status LED that lights when the relay is active, and a flyback diode already soldered on the board. Electrically identical to the standalone LU-5-R.

Coil specifications: Operating voltage = 9V DC | Coil resistance = 125 Ω | Operating current = 72 mA

Contact types — NO / NC / COM: NO (Normally Open) = open when coil is off, closes when coil is energised. NC (Normally Closed) = closed when coil is off, opens when energised. COM = common terminal. This project uses NO + COM.

Contact rating: 3A / 125V AC or 3A / 24V DC. For school projects, only low-voltage DC loads should be connected.

Why a relay and not just the transistor: The relay provides galvanic isolation — the control circuit (transistor side) and the load circuit (buzzer/LED side) are electrically separate. Also allows driving higher current loads than the transistor alone.

Flyback Diode — 1N4148

Transient protection diode | Placed across relay coil | Mandatory — not optional

What it is: A fast-switching silicon signal diode. It conducts current in one direction only (from anode to cathode). During normal operation it is reverse-biased and invisible to the circuit.

The problem it solves: When the transistor switches OFF, the magnetic energy stored in the relay coil has nowhere to go. By Faraday's law, the collapsing magnetic field induces a voltage spike. Without protection, this spike (theoretically ~3,600V in this circuit) would instantly destroy the transistor.

How it works: The diode is connected in reverse-parallel across the coil (cathode to 9V, anode to collector). The spike appears as a reverse voltage across the diode — which immediately forward-biases it, creating a safe path for the coil current to recirculate and dissipate as heat.

Clamped voltage: With the diode, the maximum voltage at the collector is clamped to $V_{supply} + V_f = 9 + 0.7 = 9.7V$. The 2N2222 withstands up to 40V — so it is fully protected.

Installation rule: Cathode (banded end) must face 9V. Anode faces the transistor collector. If installed backwards, the diode short-circuits the 9V supply — causing immediate heat damage.

Note on HW-482 module: The HW-482 module has a flyback diode already soldered on the PCB. When using HW-482, the external 1N4148 in this schematic is redundant but harmless.

RGB LED Module

Common cathode | Internal current-limiting resistors | Analog brightness control

What it is: Three LEDs (Red, Green, Blue) in one package sharing a common cathode (negative) terminal. Each colour channel has an independent anode pin. Internal resistors mean no external current-limiting resistors are needed.

RED pin — LDR1 analog control: Connected directly to LDR1 midpoint (Node A). In darkness, Node A ≈ 8.24V → RED is bright. In bright light, Node A ≈ 0.43V → RED is essentially off. Brightness varies smoothly and continuously.

GREEN pin — LDR2 analog control: Connected directly to LDR2 midpoint (Node B). Same behaviour as RED but independently controlled by LDR2. No transistor or relay in this channel.

BLUE pin: Not used in this project.

Analog nature: Unlike a digital ON/OFF signal, the midpoint voltage of the voltage divider changes smoothly with light. This means the LED brightness tracks ambient light in real time — dimming and brightening gradually.

Important distinction: The RGB module provides visual feedback of light intensity. The relay/buzzer provides a threshold-based alert (ON or OFF). Both run simultaneously from the same LDR1 signal.

Active Buzzer

5–12V DC | Internal oscillator | Driven via relay NO contact

What it is: An active buzzer contains an internal oscillator circuit. It produces a fixed-frequency tone automatically whenever DC voltage is applied — unlike a passive buzzer which requires an external oscillating signal.

How it is connected: The buzzer is connected between the relay NO contact and GND. When the relay energises (darkness on LDR1), the NO contact closes, completing the circuit to 9V, and the buzzer sounds.

Why relay-driven and not transistor-driven: The relay provides clean galvanic isolation. The buzzer's inductive load characteristics could cause interference if driven directly through the transistor. The relay's mechanical contact avoids this.

Behaviour in this circuit: Buzzer sounds in LOW light / darkness (when LDR1 resistance is high → V_mid is high → transistor saturates → relay closes). Buzzer is silent in bright light.

Key spec to check: Ensure the buzzer's rated voltage includes 9V (typically 5–12V active buzzers work at 9V). Polarity matters — positive terminal to NO contact, negative to GND.

3. Circuit Logic

3.1 LDR1 Channel — Relay / Buzzer / Red LED

Signal path

9V → 10kΩ → Node A → LDR1 → GND Node A → 1kΩ → Base (2N2222) → Relay coil → Flyback diode Relay NO → Buzzer (+) and Red LED (+) Node A → RGB RED pin (analog)

Light level	Node A voltage	Transistor	Relay / Buzzer / LED
Darkness	≈ 8.24 V	Saturated (ON)	Relay ON → Buzzer sounds, LED lit
Moderate light	~4.5 V	Partially on	Relay may be on
Bright light	≈ 0.43 V	Cut-off (OFF)	Relay OFF → Silent, LED off

3.2 LDR2 Channel — Green LED (Analog Only)

> **Signal path**
> 9V → 10kΩ → Node B → LDR2 → GND Node B → RGB GREEN pin (direct analog connection — no transistor, no relay)

LDR2 is a purely analog channel. There is no threshold switching, no transistor, and no relay. The green LED brightness varies continuously and smoothly in proportion to the light falling on LDR2. In darkness the green LED glows brightly; in bright light it is essentially off.

3.3 Simultaneous Operation

Both channels are electrically independent and operate at the same time. The table below summarises all possible combined states:

LDR1 (light level)	LDR2 (light level)	Relay / Buzzer / RED	RGB GREEN
Dark	Dark	ON	Bright
Dark	Bright	ON	Off
Bright	Dark	OFF	Bright
Bright	Bright	OFF	Off

4. Connection Diagram (Pin-by-Pin)

4.1 LDR1 Channel

- 9V (+) → 10kΩ resistor pin 1
- 10kΩ resistor pin 2 → Node A
- Node A → LDR1 pin 1
- LDR1 pin 2 → GND
- Node A → 1kΩ base resistor pin 1

- 1kΩ base resistor pin 2 → 2N2222 BASE (middle pin, flat face toward you)
- 2N2222 EMITTER (left pin) → GND
- 2N2222 COLLECTOR (right pin) → Relay coil (−) / HW-482 IN pin
- Relay coil (+) / HW-482 VCC → 9V
- 1N4148 ANODE → Relay coil (−) | 1N4148 CATHODE (banded end) → 9V
- Relay NO contact → Buzzer (+) and Red LED anode
- Buzzer (−) → GND | Red LED cathode → GND
- Node A → RGB LED RED pin

4.2 LDR2 Channel

- 9V (+) → 10kΩ resistor pin 1
- 10kΩ resistor pin 2 → Node B
- Node B → LDR2 pin 1
- LDR2 pin 2 → GND
- Node B → RGB LED GREEN pin

4.3 Power

- 9V battery (+) → breadboard positive rail
- 9V battery (−) → breadboard GND rail

5. Troubleshooting

Symptom	Likely cause	Fix
Relay never activates even in darkness	LDR and resistor swapped (v1 wiring)	Verify: 9V → 10kΩ → Node → LDR → GND
Relay always ON even in bright light	Transistor shorted or B/C/E pins reversed	Check 2N2222 pinout; replace if shorted
LED doesn't respond to LDR	Node not connected to RGB pin	Check jumper from divider midpoint to RGB pin
Relay chatters (clicking repeatedly)	V_mid oscillating near threshold	Add 100nF cap from base to GND
Transistor gets hot	Not in full saturation	Reduce base resistor value to increase I_B
No output at all	Battery flat or rail not connected	Measure 9V supply; check polarity

6. Safety Notes

- Always disconnect the battery before modifying any wiring.
- Verify 2N2222 pin orientation before powering up (EBC from flat face — left to right).
- 1N4148 flyback diode: cathode (banded end) must face 9V. Reversed = short circuit.
- HW-482 module already includes a flyback diode on-board. External 1N4148 is redundant but safe to keep.
- Relay contact rated 3A / 24V DC maximum. Never connect mains voltage in a school project.
- Short circuits at 9V will not cause electric shock but will rapidly heat wires and damage components.
- If any component becomes hot, disconnect the battery immediately.